

Field system board replacement procedure

System board replacement procedure and flow chart

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SE350 system board replacement procedure

The SE350 system board replacement procedure is unique due to the need for the servicer to upload the new Activation Code to the replacement system board, and, if SED is used, to restore the SED AK from the SED AK backup file.

Attention: This information is for ThinkShield Service Support users only. Anyone servicing an SE350 should send an email to thinkshield@lenovo.com to have a ThinkShield Service Support user account created.

Click each number in turn to see the procedure.

Step

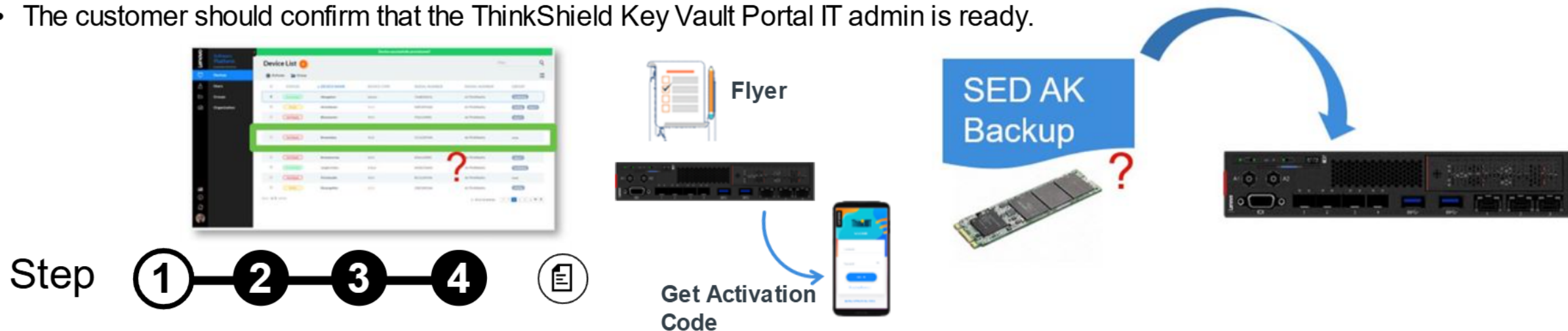


SE350 system board replacement procedure

- Make sure ThinkShield Service Support users have a ThinkShield Service Support user account. They can create an account by sending an email to thinkshield@lenovo.com

Note: The customer is responsible for the following steps.

- Confirm the customer has claimed and activated the SE350 through the ThinkShield Key Vault Portal.
 - Refer to the **Claiming the device** section in this course for details.
 - Confirm the customer has the activation flyer for the Activation Code.
 - If they do not have the flyer, retrieve the Activation Code through the mobile app.
- If SED is used, the customer needs to confirm there is an SED AK backup file. Refer to the **Backup the SED AK** section in this course for details.
 - If the customer does not have a backup file, the data cannot be recovered.
- The customer should confirm that the ThinkShield Key Vault Portal IT admin is ready.



SE350 system board replacement procedure

Note: The servicer is responsible for the following steps.

- Back up the machine type, serial number, UUID (if needed), and UEFI / XCC settings from XCC.
- Turn off the system and remove the ac power source.
- Open the top cover and record the location of all the components mounted on the system board. This will include the CPU, DIMMs, power/signal cables, risers, adapters, and M.2 adapter.
- Remove all the components mounted on the system board.
- Remove the failure system board from the chassis.
- Install the new system board in the chassis.
- Install all the components mounted on the system board.
- Connect the power source. Do not turn on the system at this stage.
- Restore the machine type, serial number, UUID (if needed) by using OneCLI. Refer to the following Web page for detailed instructions: [How to update VPD using OneCLI](#)

Step



SE350 system board replacement procedure

Attention: If the **servicer** cannot access the mobile app, the **customer** should contact thinkshield@lenovo.com to manually update the Activation Code.

When Lenovo service personnel replace a system board, the board will have a new Activation Code but the ThinkShield Key Vault Portal still has the old Activation Code.

- Lenovo service personnel can replace the Device key with the Mobile application. Customers in Maintenance User roles can replace the Device key using Lenovo Service Utility for ThinkShield. Customers can get the Device key from the QR code on the new system board or through IPMI and then ask a Maintenance User to work with the Org Admin to manually replace it through the Portal.
 - Mobile application: refer to the [Checking and updating the Device key](#) section of this course for more information.
 - Lenovo Service Utility for ThinkShield: refer to the [Lenovo Service Utility for ThinkShield](#) section of this course for more information.
 - IPMI: refer to the [How to use IPMI to get a Device key](#) section.
 - The user manually replaces the Device key: Refer to the [ThinkShield Web application - Update key](#) section of this course for more information.
- The Device key will update the Activation Code for the system board. Record the new Activation Code for your records.
 - The new system board will have a new Secure Activation Code on a sticker on the heatsink. Service personnel will need to put the new sticker on the SE350 pull-out asset tag.
- If the SE350 has been claimed, the SE350 will be available under the customer organization in the ThinkShield Key Vault Portal.
 - If the SE350 has not been claimed, it needs to be claimed for it to show up in the ThinkShield Key Vault Portal. Refer to the [Claiming the device](#) section in this course for details of how to claim an SE350.
 - If the activation flyer is available, or if the Activation Code can be accessed from the old motherboard using the mobile app, the customer IT admin can claim the SE350.
 - If not, Lenovo service personnel needs to contact the ThinkShield Key Vault Portal support desk to claim the SE350 by MT / SN (without the Activation Code).

Step



SE350 system board replacement procedure

Note: The customer is responsible for the following steps.

- Log in to XCC and restore FoD key(s) and the SED AK from the SED AK backup file.
 - If the customer uses SED but does not have an SED AK backup file, the data cannot be recovered.

Note: The servicer is responsible for the following steps.

- Turn on the system and open the F1 UEFI setup menu. Make sure the CPU and DIMM inventory in the F1 UEFI menu is correct.
- Start the OS and make sure every function is working in accordance with the customer's expectation.

Step  

PSID revert overview

If the customer has lost the SED AK, we cannot recover the content. However, by running the Physical Security ID (PSID) revert process, the customer can uninstall the SED AK to repurpose the media. This process will also destroy the SED content.

It is worthwhile knowing the PSID revert process, but it is not the servicer's responsibility to execute it. It is customer's responsibility to maintain the backup SED AK.

There are two tools for the PSID revert process:

- Micron media (`msecli`) This is a publically available tool from Micron, and detailed specifications are available from Micron:
 - https://www.micron.com/-/media/client/global/documents/products/software/storage-executive-software/msecli/storage_executive_cli_user_guide.pdf?la=en
 - Refer to the **Performing a PSID Revert (SATA and SAS Drives Only)** section for detail instructions.
- Other media (Intel, ATP) uses an open source tool called `sedutil-cli`. Information about the tool can be found at the following Web site:
 - <https://github.com/Drive-Trust-Alliance/sedutil/wiki/PSID-Revert>

PSID revert process

To execute the PSID revert process, follow the steps set out below.

- Prepare the unlocked boot media
- Install Linux, [sedutil](#) (open source) and the [Micron tool](#)
- Record each media PSID
- Attach the SED to the SE350 M.2 adapter (SATA / NVMe)
- Revert the PSID (to uninstall the SED AK from the media)
 - With Micron SATA
 - Run the micron tool showing PSID
 - With Intel / ATP
 - Run sedutil showing PSID
- Reverting the PSID will also destroy the SED content.



Record each PSID of media

PSID revert CLI command – Micron tool

Micron tool syntax:

```
msecli -X -P -p <PSID-value> -n <device-name>
```

Where:

-X = secure erase/sanitize/PSID option

-P = PSID operation

-p = the drive's PSID value which can be found on the drive's front label

-n = specifies which drive you are performing the operation on

Sample usage:

```
C:\Users\Administrator>msecli -X -P -p 01234567890123456789012345678901 -n Drive0
PSID Revert will permanently erase ALL user data! Please backup all data from the drive before proceeding!
Are you sure you want to continue(Y|N):y
PSID Revert will take a few seconds to complete.
...
Device Name   : Drive0
PSID Revert completed successfully
CMD_STATUS    : Success
STATUS_CODE   : 0

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C:\Users\Administrator>
```

PSID revert CLI command – sedutil

sedutil tool syntax:

```
sedutil-cli --yesIreallywanttoERASEALLmydatausingthePSID < PSID-value> <device-name>
```

Sample usage:

```
./sedutil-cli --yesIreallywanttoERASEALLmydatausingthePSID 0123456789987654321  
/dev/sda
```

The following information will show if the PSID successfully reverts:

```
revertTper completed successfully.
```

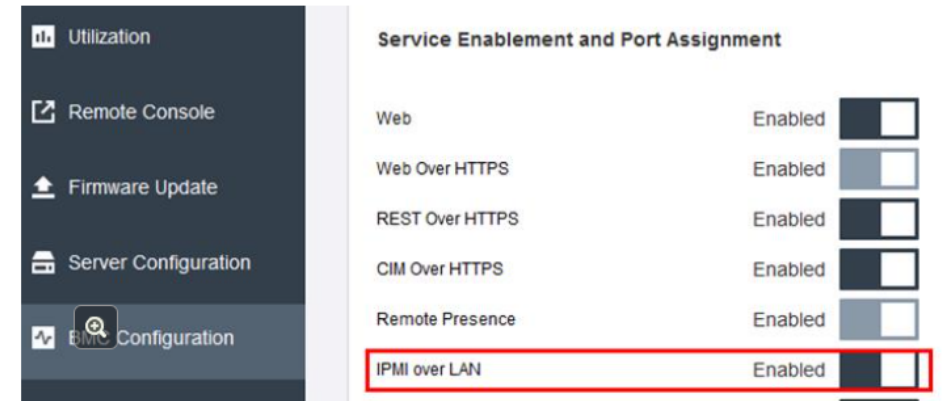
If you get a NOT_AUTHORIZED message, it means the PSID you entered was wrong.

Sedutil usage reference [Web page](#).

How to use IPMI to get a Device key

Enable IPMI in XCC:

- Select **BMC Configuration** → **Network**, and scroll down to the **Service Enablement and Port Assignment**.
- Configure the state of the **IPMI over LAN** to **Enabled**.



Connect the laptop to the XCC port that has IPMITool installed. Assume the XCC IP address is 192.168.70.125, the user ID is USERID, and the password is PASSWORD (zero, not capital o). Use the following command to get the Device key:

```
>ipmitool -I lanplus -U USERID -P PASSWORD -H 192.168.70.125 raw 0x3a 0xf2 0x6
```

Sample output:

```
54 41 8b d7 76 4f 1b 57 bf 63 0e 6b 31 ea 27 98
0b a4 29 e1 b2 12 86 c2 42 70 aa 2d 17 f7 26 ce
88 0c a8 a8 cb e9 0c cd 72 d2 74 05 f7 8b 7d a0
4f f9 3f 9e da 95 8f 1e a8 85 25 10 9a bd fe 9e
```

The SE350 Device key will be (64 bytes, 128 hex chars long):

```
54418bd7764f1b57bf630e6b31ea27980ba429e1b21286c24270aa2d17f726ce880ca8a8cbe90ccd72d27405
f78b7da04ff93f9eda958f1ea88525109abdfe9e
```

How to use IPMI to get an activation code (AC)

Connect the laptop to the XCC port that has IPMITool installed. Assume the XCC IP address is 192.168.70.125, the user ID is USERID, and the password is PASSWORD (zero, not capital o). Use the following command to get an activation code:

```
>ipmitool -I lanplus -U USERID -P PASSWORD -H 192.168.70.125 raw 0x3a 0xf2 0x10 0x03
```

The out put will be 16 bytes, and the hex value, 16 characters.

Sample output:

```
34 43 37 44 32 36 38 32 31 33 46 32 30 39 37 34
```

Using the online [Hex To String Convert](#) tool to transform the hexadecimal text to a decoded string, the output will be:

```
4C7D268213F20974
```

The Activation Code on the label will be **4C7D-2682-13F2-0974**

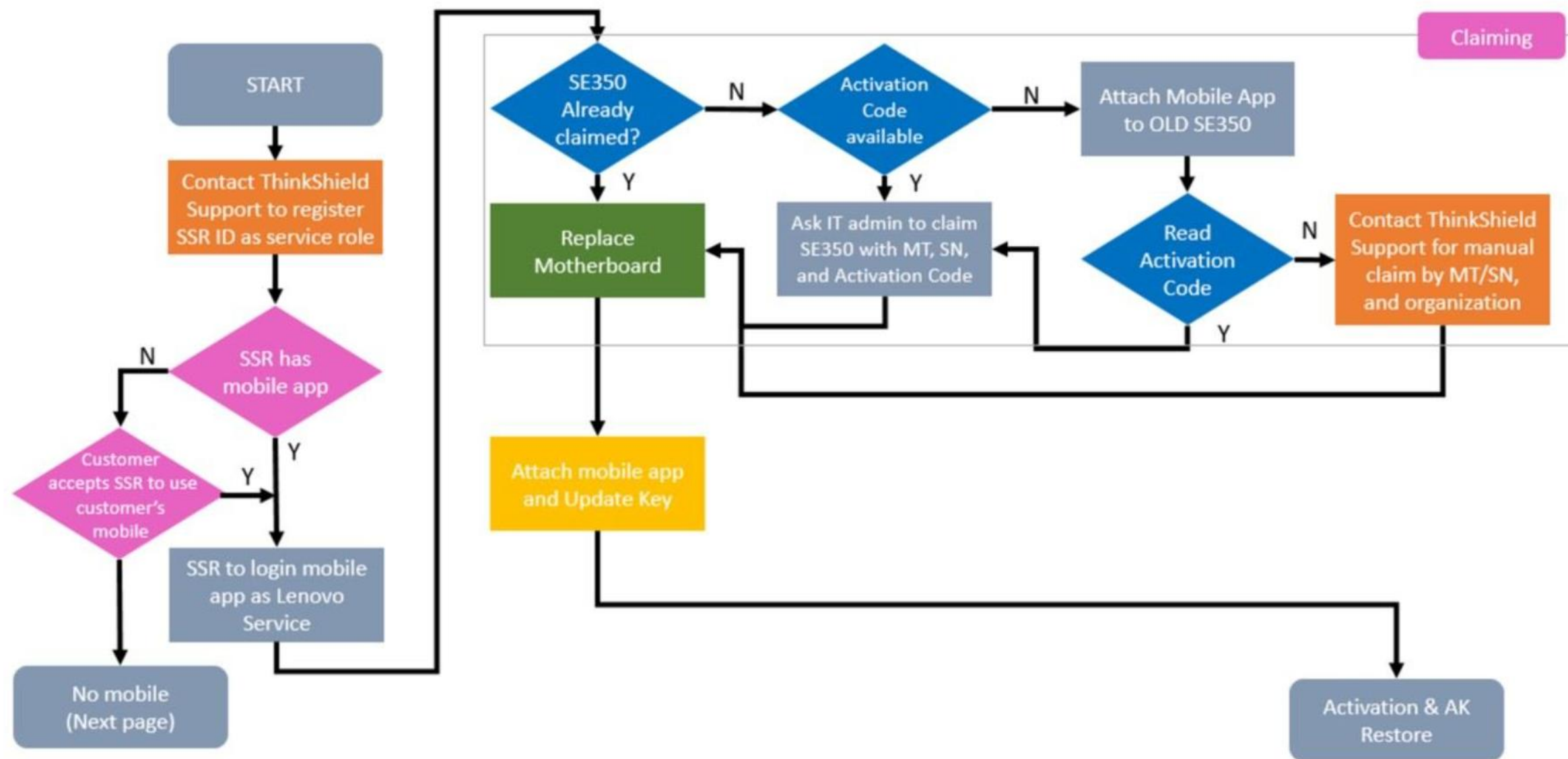
How to use IPMI to set the device mode as Security Pack

Connect the laptop to the XCC port that has IPMITool installed, and then run the following command to set the device mode as Security Pack:

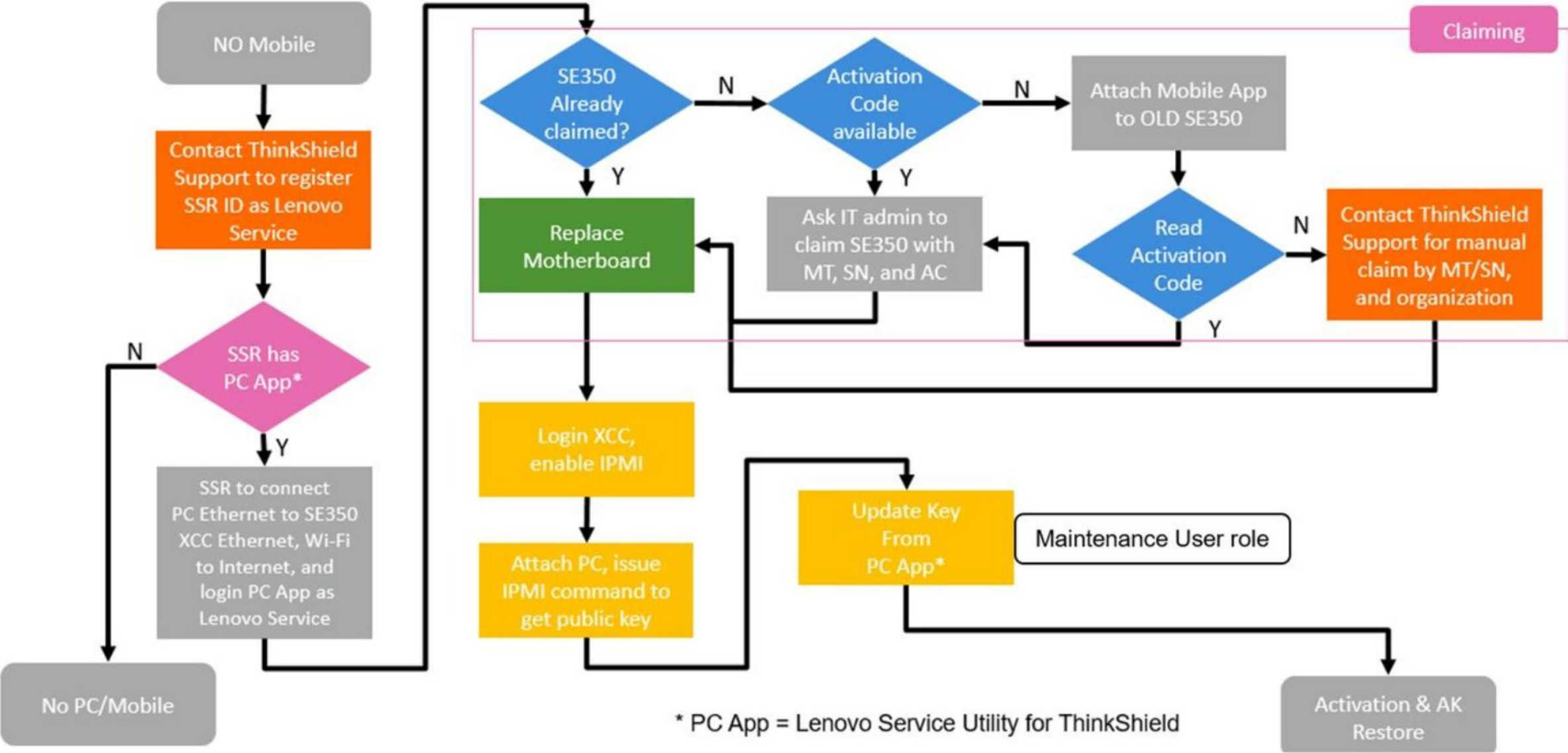
```
>ipmitool -I lanplus -U USERID -P PASSWORD -H 192.168.70.125 raw 0x3a 0xf2 0x10 0xea 0x1
```

Note: In this example, the XCC IP address is 192.168.70.125, the user ID is USERID, and the password is PASSWORD (zero, not capital o).

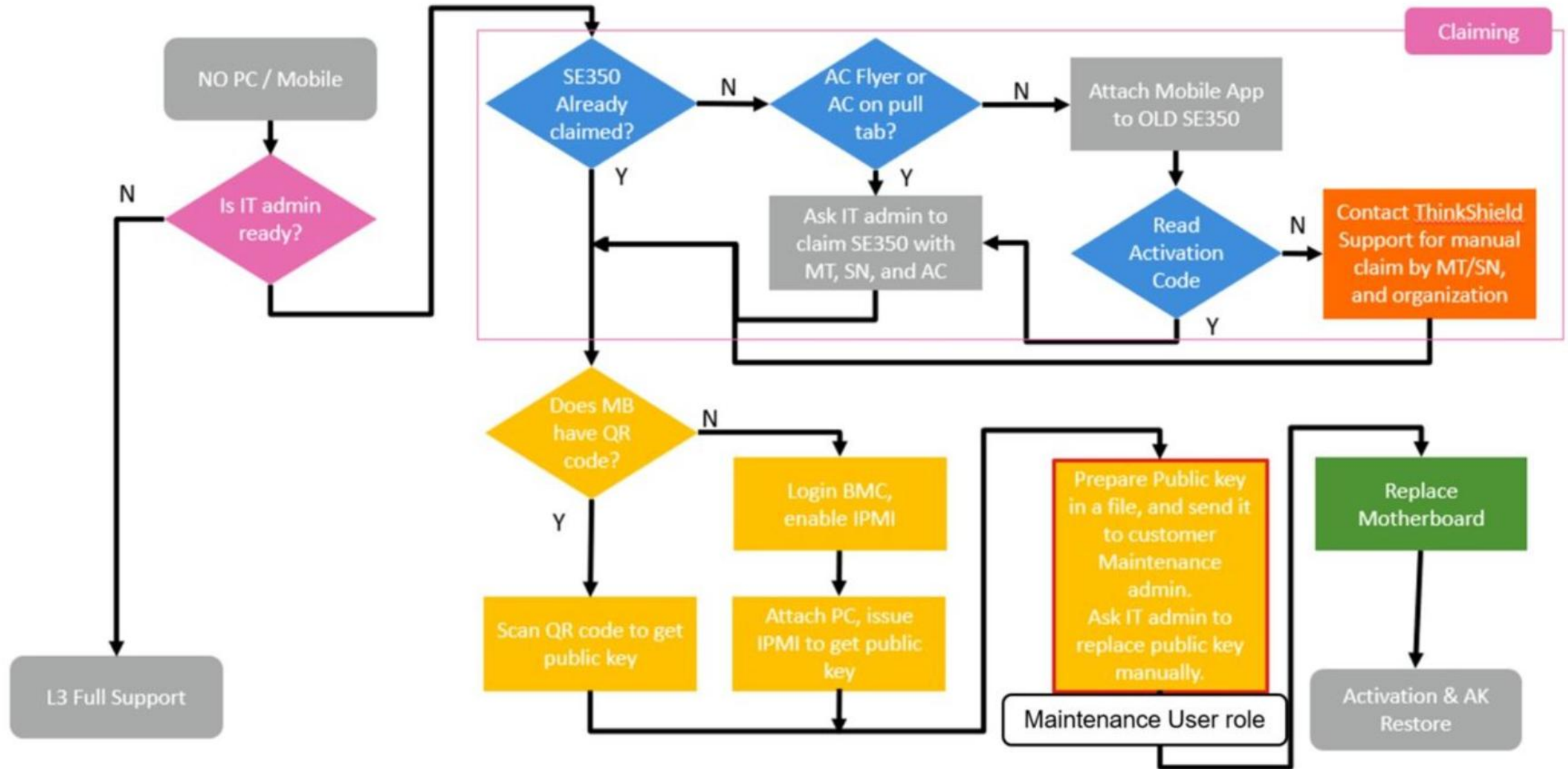
System board replacement flow chart – suitable for mobile app use



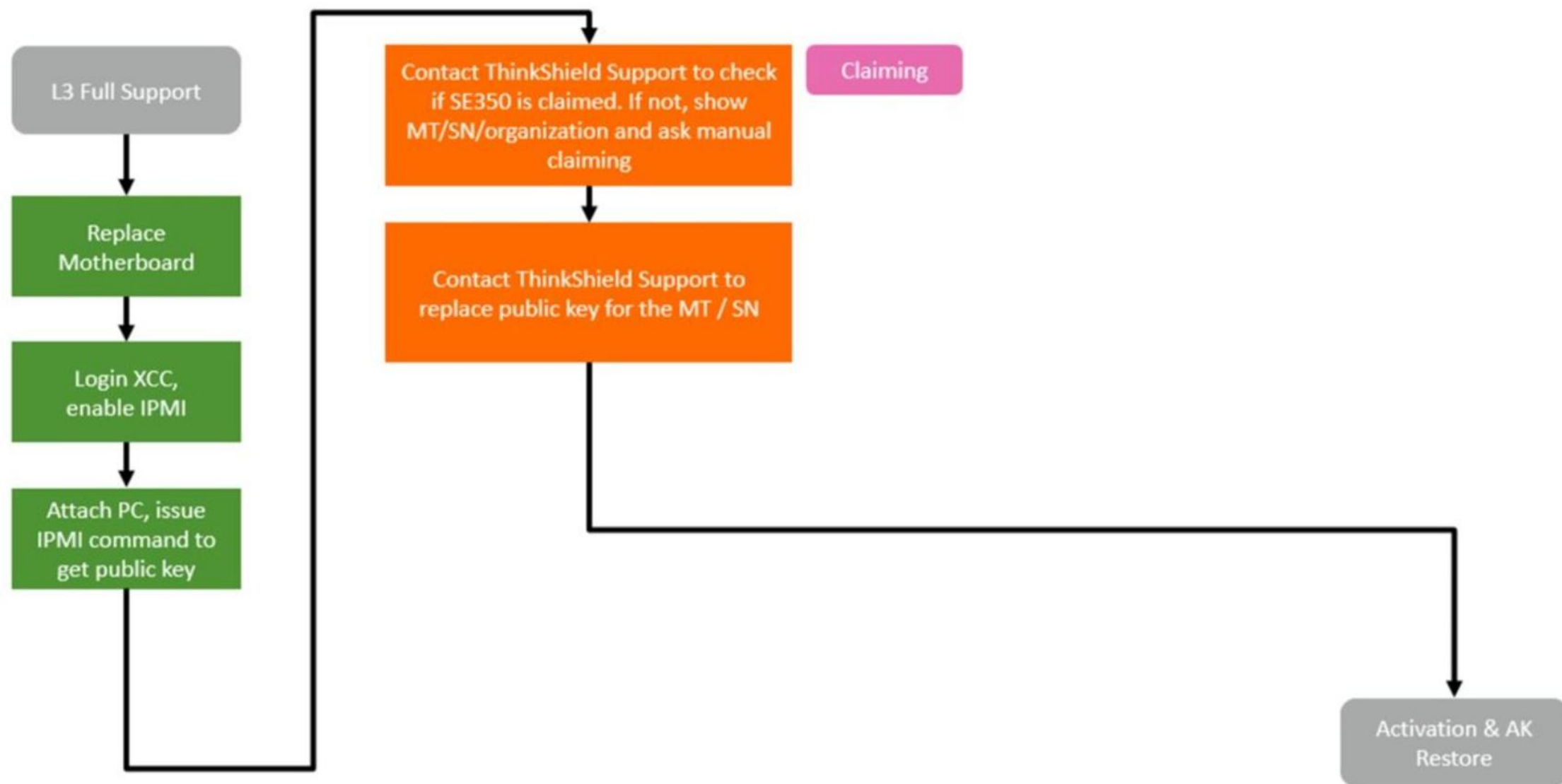
System board replacement flow chart – suitable for PC use



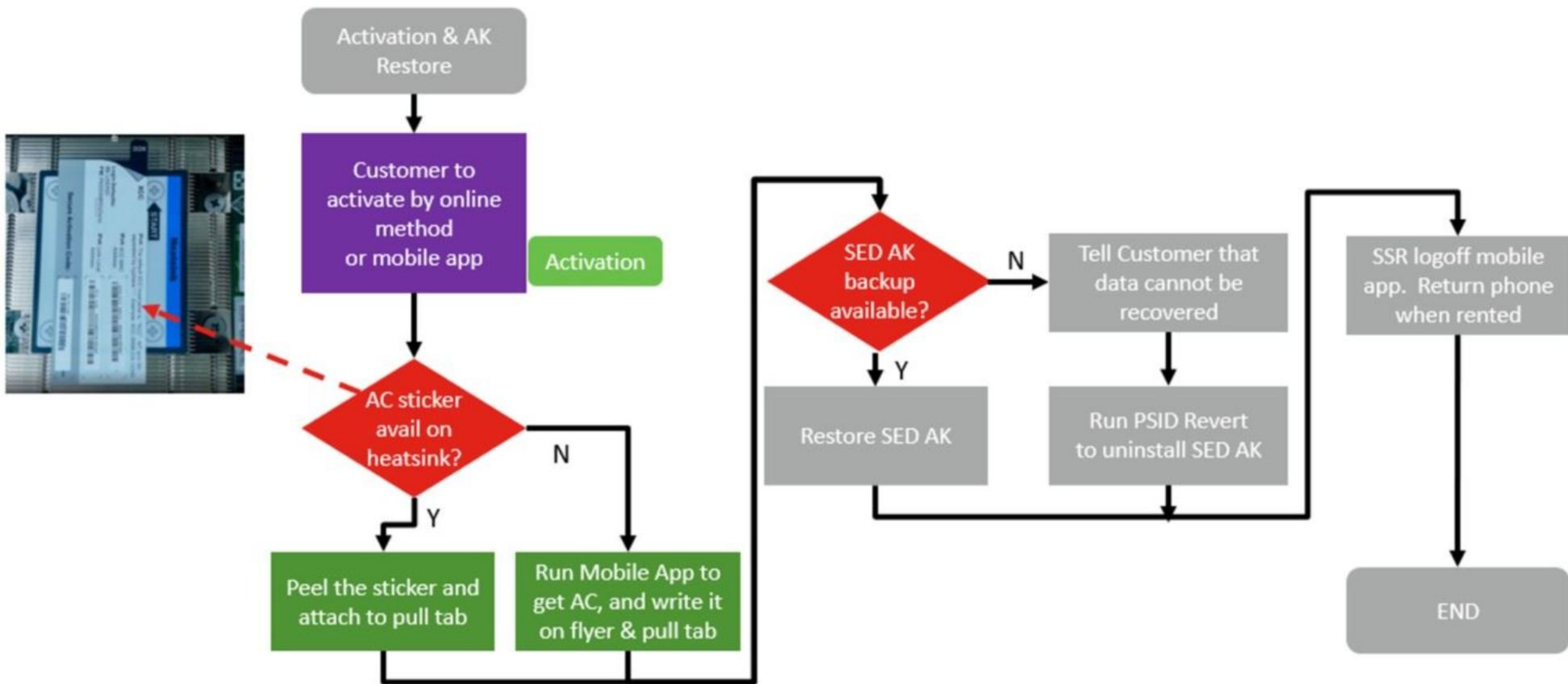
System board replacement flow chart – IT Admin support



System board replacement flow chart – L3 full support (no mobile app)



Activation and SED AK restoration flow chart



Customer responsibilities

Identifying the responsibilities of the customer

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Customer responsibilities

To help servicers, the customer is responsible for the following items:

- Keeping the Activation Code (flyer)
- Maintaining a backup of the SED AK
- Moving the SE350 to a safe working place for the field service
- Preparing the customer's mobile phone cable
- Engaging the IT department so that they can help to claim or activate the device when required
 - Confirming whether the HW has been claimed or not. If it has not been claimed, working with the IT department to claim the device.
- Restoring the SED AK and the password from the backup file
- Returning the SE350 to its usual location
- Ensuring wireless (network) connectivity
- Restoring FoD after a system board has been replaced
- Maintaining a backup of the FoD key(s)